

Connected Warehouse Vision Proof of Concept

Prepared for Harbor & Vale Logistics by Solstice Systems · Field Solutions

1 / GOAL

Prove that Solstice can detect blocked dock doors and forklift near-misses across three distribution centers within two seconds of the event, route alerts to the operations team automatically, and give the analytics team a queryable history of every event, all managed as a single fleet from one console.

2 / PRODUCTS IN SCOPE

PRODUCT	ENVIRONMENT	EFFORT
Solstice Core Platform	New deployment	20h
Outpost Edge Runtime	New deployment	20h
Outpost Fleet Manager	New deployment	17h
Solstice Streams (required foundation)	New deployment	15h
Solstice Lakehouse (required foundation)	New deployment	17h
Nova Insight	New deployment	22h
Nova Vision	New deployment	29h
Conductor Workflows	New deployment	10h

Solstice Core Platform

Control plane footprint: Highly available (3 nodes)	6h
Target platform: VMware vSphere	4h
Network connectivity: Through a corporate proxy	2h
Persistent storage: SAN / NAS via CSI driver	3h
Platform identity: OIDC single sign-on	3h
Integrate with Harbor & Vale's Okta tenant for platform sign-in	2h

Outpost Edge Runtime

Base work package	6h
Edge sites in scope: 3 sites	4h
Site hardware: Vendor-certified gateways	2h
Provisioning: Zero-touch provisioning	5h
Site connectivity: Intermittent with store-and-forward	3h

Outpost Fleet Manager

Base work package	8h
Management scope: Configuration + over-the-air updates	5h
Inventory integration: Sync with existing CMDB / ITSM	4h

Solstice Streams

Base work package	6h
Broker topology: 3 brokers	3h
Retention: Tiered to object storage	3h
Source connectors: Files and MQTT	3h

Solstice Lakehouse

Base work package	10h
Object storage: Bring your own S3-compatible store	2h
Query compute: Dedicated query cluster	4h
Table catalog: Built-in catalog	1h

Nova Insight

Base work package	10h
Workload: Training + model serving	8h
Accelerators: GPU pool on Core workers	4h

Nova Vision

Base work package	12h
Camera count: 10 cameras	4h
Detection models: Custom model training in Nova Insight	8h
Inference target: Edge GPU	4h
Mask the customer-facing lobby camera region for privacy	1h

Conductor Workflows

Base work package	6h
Engine deployment: Shared engine on Core	1h
Integrations: ITSM ticketing	3h

4 / PREREQUISITES

Solstice Core Platform

CUSTOMER PROVIDES	VALUE	STATUS
Control plane VIP	10.42.0.10	Provided
Application ingress VIP	10.42.0.11	Provided
Node network	10.42.0.0/24	Provided
DNS records	core.poc.harborvale.example, *.apps.poc.harborvale.example	Provided
NTP source	ntp.harborvale.example	Provided
SSH public key	Provided by platform team	Provided
Solstice license key	—	Pending
Out-of-band management (optional)	—	Not applicable

Outpost Edge Runtime

CUSTOMER PROVIDES	VALUE	STATUS
Site list	Newark DC, Savannah DC, Tacoma cross-dock	Provided
Gateway hardware specs	Outpost G2 gateway, 16 GB, 1 TB NVMe, T4 GPU	Provided
Site network details	VLAN 220, DHCP reservations, outbound 443 only	Provided
Site escort for gateway installation	Facilities to confirm per site	Pending

Outpost Fleet Manager

CUSTOMER PROVIDES	VALUE	STATUS
CMDB / ITSM API access (optional)	—	Pending
Maintenance window per site	Sundays 02:00–04:00 local	Provided

Solstice Streams

CUSTOMER PROVIDES	VALUE	STATUS
Initial topic list	dock.events, forklift.telemetry, camera.alerts	Provided
Source system access (optional)	—	Pending

Solstice Lakehouse

CUSTOMER PROVIDES	VALUE	STATUS
Object storage bucket	s3://harborvale-poc (MinIO)	Provided
Sample data set	—	Pending

Nova Insight

CUSTOMER PROVIDES	VALUE	STATUS
GPU nodes (optional)	2 × NVIDIA L4 on Core workers	Provided
Data owner sign-off	—	Pending

Nova Vision

CUSTOMER PROVIDES	VALUE	STATUS
Camera inventory	10 Axis P3265 dome cameras, RTSP, dock doors 1–10	Provided
Labeled sample footage (optional)	—	Pending
Privacy review	Approved by Legal 2026-09-08	Provided

Conductor Workflows

CUSTOMER PROVIDES	VALUE	STATUS
Integration service account (optional)	—	Pending
First three workflows to automate	Dock-door blocked alert, forklift near-miss report, nightly site health summary	Provided

5 / DELIVERY PLAN

1. Engagement Planning

1.1. Kickoff, access validation, and success criteria (Planning)	3h
1.2. Target architecture and site plan (Planning)	4h
1.3. Executive readout and handover (Testing and Validation)	3h

2. Core Platform Operations

2.1. Enable monitoring, alerting, and log forwarding (Configuration)	3h
2.2. Configure control plane backup and restore drill (Configuration)	4h

4. Automation Workflows

4.1. Build the first three workflows (Configuration)	6h
4.3. Run workflows end to end with the operations team (Testing and Validation)	3h

5. Streaming Pipeline

5.1. Create topics, schemas, and access controls (Configuration)	3h
5.2. Wire source connectors and validate throughput (Testing and Validation)	4h

6. Lakehouse Analytics

6.1. Land streaming data as open tables (Configuration)	4h
6.2. Build three business dashboards (Configuration)	5h

7. Edge Site Enablement

7.1. Build the site image and provisioning profile (Deployment)	4h
7.2. Bring up the first site and validate telemetry (Deployment)	4h
7.3. Validate store-and-forward during a link outage (Testing and Validation)	3h

8. Fleet Operations

8.1. Define site groups and configuration baselines (Configuration)	3h
8.2. Roll an over-the-air update to a canary group (Testing and Validation)	4h

9. Model Development

9.1. Provision workbenches and connect Lakehouse data (Configuration)	3h
9.2. Train a baseline model on the sample data (Configuration)	6h
9.3. Serve the model and validate latency (Testing and Validation)	4h

10. Vision Use Case

10.1. Onboard pilot cameras and validate ingest (Deployment)	3h
10.2. Tune detection for the customer use case (Configuration)	6h
10.3. Route alerts through Conductor to operations (Configuration)	3h
10.4. Acceptance test against agreed detection targets (Testing and Validation)	4h

6 / EFFORT SUMMARY

Installation and configuration	150h
Deliverables	89h
Total estimated effort	239h
Planning range	192–287h

7 / SUCCESS CRITERIA

Blocked dock-door events are detected at all three sites within two seconds, with fewer than one false alert per site per day.

Forklift near-miss alerts reach the operations channel through Conductor with the camera clip attached.

Every event is queryable in Lakehouse dashboards within one minute of occurring.

A configuration change and an over-the-air update roll to all three sites from Fleet Manager with staged canary and rollback.

8 / ASSUMPTIONS

Harbor & Vale provides VMware capacity, network access, and camera credentials before kickoff.

The three pilot sites have a technician available for gateway installation.

Sample footage for custom training is delivered by the end of week one.
Estimate covers Solstice delivery effort only; calendar duration depends on site access and change windows.

CUSTOMER	ACCOUNT TEAM	FIELD SOLUTIONS
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